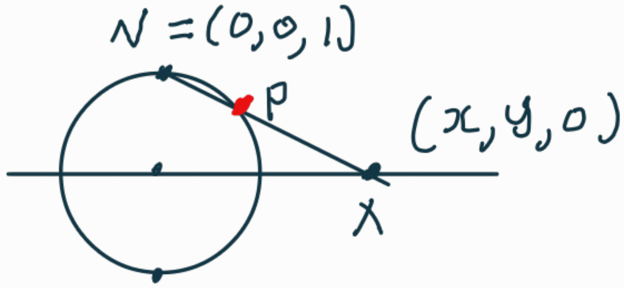
Round metric and the curvature



$$P = N + t(X - N) = (1-t)N + tX$$

Since P is on S^2,

$$(tx,\ ty,\ 1-t) \in S^2$$

$$\overset{t\neq 0}{\Longrightarrow} \quad t^2(x^2+y^2) + t^2 - 2t + 1 = 1$$

$$t(x^2+y^2) + t = 2$$

$$t = \frac{2}{x^2+y^2+1}$$

So, $$P = \left(\frac{2x}{x^2+y^2+1},\ \frac{2y}{x^2+y^2+1},\ 1 - \frac{2}{x^2+y^2+1} \right)$$

We can compute first fundamental form as follows.

$$P_x = \left(\frac{2(x^2+y^2+1) - 4x^2}{(x^2+y^2+1)^2},\ \frac{2y\cdot(-2x)}{(x^2+y^2+1)^2},\ \frac{4x}{(x^2+y^2+1)^2} \right)$$

$$P_y = \left(\frac{2x\cdot(-2y)}{(x^2+y^2+1)^2},\ \frac{2(x^2+y^2+1)-4y^2}{(x^2+y^2+1)^2},\ \frac{4y}{(x^2+y^2+1)^2} \right)$$

$$\langle P_x, P_y \rangle$$

$$= \frac{2(-x^2+y^2+1)\cdot(-4xy)+2(x^2-y^2+1)(-4xy)+16xy}{(x^2+y^2+1)^4}$$

$$= \frac{4\cdot(-4xy)+16xy}{(x^2+y^2+1)^4}$$

$$= 0.$$

$$\langle P_x, P_x \rangle = \frac{1}{(x^2+y^2+1)^4}\left((2(x^2+y^2)+2-4x^2)^2 + 16x^2y^2 + 16x^2 \right)$$

$$= \frac{1}{(x^2+y^2+1)^4}\left((2(x^2+y^2)+2) - (16S+16)x^2 + \underbrace{16x^4+16x^2y^2+16x^2}_{=16x^2(x^2+y^2+1)} \right)$$

$$= \frac{1}{(x^2+y^2+1)^4}\left(\underbrace{4(x^2+y^2)^2+4(x^2+y^2)+4}_{=4(x^2+y^2+1)^2} - 16x^2 + 16x^2 \right)$$

$$= \frac{4}{(x^2+y^2+1)^2}. \qquad \therefore \langle P_y, P_y \rangle = \langle P_x, P_x \rangle$$

$$, \ \langle P_y, P_y \rangle = \frac{4}{(S+1)^2}.$$

So, $$ds^2 = \frac{4}{(x^2+y^2+1)^2}(dx^2+dy^2).$$

First observation: The gauss curvature $= 1$.

$P(x, y)$ is the unit normal vector of the unit sphere.

So, the shape operator is given by

$$-dN\left(\frac{\partial P}{\partial x}\right) = -N_x = -\left(\frac{2(x^2+y^2+1)-4x^2}{(x^2+y^2+1)^2}, \frac{2y\cdot(-2x)}{(x^2+y^2+1)^2}, \frac{4x}{(x^2+y^2+1)^2}\right)$$

A tangent vector of S^2

$$= -\frac{\partial P}{\partial x}$$

$$-dN\left(\frac{\partial P}{\partial y}\right) = -\frac{\partial P}{\partial y}$$

So, the determinant of the shape operator $= 1$.

$\boxed{\text{III}}$ - Derivation of the formula from shape op..

$$\boxed{K = -\frac{1}{2e^{\sigma}}\Delta\sigma, \text{ where } \sigma = \log(\lambda)}$$

$P : U \xrightarrow{\subseteq \mathbb{R}^2} \Sigma$ = surface is a coordinate patch

And let n be the normal vector.

$$E = \langle P_x, P_x \rangle, \quad F = \langle P_x, P_y \rangle, \quad G = \langle P_y, P_y \rangle$$

$$L = \langle P_{xx}, n \rangle, \quad M = \langle P_{xy}, n \rangle, \quad N = \langle P_{yy}, n \rangle.$$

For example if we take Σ as S^2, the shape operator W is just

$$W(\alpha P_x + \beta P_y) \quad \left(:= \left(-dn \left(\alpha \frac{\partial P}{\partial x} + \beta \frac{\partial P}{\partial y} \right) \right) = \frac{d}{dt} n \circ \gamma \quad \gamma' = (r_1', r_2', r_3') = \alpha \frac{\partial P}{\partial x} + \beta \frac{\partial P}{\partial y} \quad = -\alpha \frac{\partial P}{\partial x} - \beta \frac{\partial P}{\partial y}. \right)$$

$$= \alpha W(P_x) + \beta W(P_y)$$

$$= \alpha n_x + \beta n_y$$

So, $\langle W(\alpha P_x + \beta P_y), \alpha P_x + \beta P_y \rangle$

$$= \alpha^2 \underbrace{\langle W(P_x), P_x \rangle}_{=L} + \alpha\beta \underbrace{\langle W(P_x), P_y \rangle}_{=M} + \alpha\beta \underbrace{\langle P_y, W(P_x) \rangle}_{=M} + \beta^2 \underbrace{\langle W(P_y), P_y \rangle}_{=N}$$

$$\left(\langle n, P_x \rangle = 0 \Rightarrow \langle n_x, P_x \rangle = \langle -W(P_x), P_x \rangle = -\langle n, P_{xx} \rangle = -L \right)$$

$$= \alpha^2 L + 2\alpha\beta M + \beta^2 N \quad \text{--- } (*)$$

Let $W(P_x) = a P_x + c P_y$

$W(P_y) = b P_x + d P_y$

$$\left(K = \det \begin{pmatrix} a & c \\ b & d \end{pmatrix} \right)$$

$\langle W(P_x), P_x \rangle = L$ by $(*)$

$$= a\langle P_x, P_x\rangle + c\langle P_y, P_x\rangle$$

$$= aE + cF$$

$\langle W(P_x), P_y \rangle = M$ by $(*)$

$$= a\langle P_x, P_y\rangle + b\langle P_y, P_y\rangle$$

$$= aF + cG$$

$$\begin{bmatrix} E & F \\ F & G \end{bmatrix} \begin{bmatrix} a \\ c \end{bmatrix} = \begin{bmatrix} L \\ M \end{bmatrix}$$

Simmilarly, $\begin{bmatrix} E & F \\ F & G \end{bmatrix} \begin{bmatrix} d \\ b \end{bmatrix} = \begin{bmatrix} F \\ M \end{bmatrix}$.

So, $\begin{bmatrix} E & F \\ F & G \end{bmatrix}^{-1} \begin{bmatrix} L & M \\ M & F \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$K = \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \frac{LN - M^2}{EG - F^2}$$

How to derive it from the first fundamental form? Consider the following matrices

$$\left[\, P_{xx} \;\middle|\; P_x \;\middle|\; P_y \,\right] \cdots (1)$$

$$\left[\, P_{xy} \;\middle|\; P_x \;\middle|\; P_y \,\right] \cdots (2)$$

$$\left[\, P_{yy} \;\middle|\; P_x \;\middle|\; P_y \,\right] \cdots (3)$$

$$\det(1) = \langle P_{xx}, P_x \times P_y \rangle = \|P_x \times P_y\| L$$

$$\det(2) = \langle P_{xy}, P_x \times P_y \rangle = \|P_x \times P_y\| M$$

$$\det(3) = \langle P_{yy}, P_x \times P_y \rangle = \|P_y \times P_y\| N$$

$$K = \frac{\det(1)\det(3) - [\det(2)]^2}{\underline{\|P_x \times P_y\|}^2 (EG - F^2)}$$

$$= \det \begin{bmatrix} \langle P_x, P_x \rangle & \langle P_x, P_y \rangle \\ \langle P_y, P_x \rangle & \langle P_y, P_y \rangle \end{bmatrix}$$

$$= \frac{\det(1)\det(3) - [\det(2)]^2}{(EG - F^2)^2}$$

$$\Rightarrow \|P_x \times P_y\|^2$$
$$= (\|P_x\| \, \|P_y\| \sin\theta)^2$$
$$= \|P_x\|^2 \|P_y\|^2$$
$$- \cos^2\theta \|P_x\| \|P_y\|$$

$\det ①③$

$$= \det \left[\begin{bmatrix} P_{xx}^T \\ \hline P_x^T \\ \hline P_y^T \end{bmatrix} \begin{bmatrix} P_{yy} & P_x & P_y \end{bmatrix} \right]$$

$$= \det \begin{bmatrix} P_{xx}^T P_{yy} & P_{xx}^T P_x & P_{xx}^T P_y \\ P_x^T P_{yy} & P_x^T P_x & P_x^T P_y \\ P_y^T P_{yy} & P_y^T P_x & P_y^T P_y \end{bmatrix}$$

$$= \det \begin{bmatrix} P_{xx}^T P_{yy} & \langle P_{xx}, P_x \rangle & \langle P_{xx}, P_y \rangle \\ \langle P_x, P_{yy} \rangle & E & F \\ \langle P_y, P_{yy} \rangle & F & G \end{bmatrix}$$

we can express the with P.P.A.

we can not express each of them with P.P.A.

$\det ②②$

$$= \det \begin{bmatrix} P_{xy}^T P_{yx} & \langle P_{xy}, P_x \rangle & \langle P_{xy}, P_y \rangle \\ \langle P_x, P_{xy} \rangle & E & F \\ \langle P_y, P_{xy} \rangle & F & G \end{bmatrix}$$

$\det ①③$

$$= P_{xx}^T P_{yy} \cdot (EG - F^2)$$

$$- \langle P_x, P_{yy} \rangle \cdot \left(\langle P_{xx}, P_x \rangle G - \langle P_{xx}, P_y \rangle F \right)$$

$$+ \langle P_y, P_{yy} \rangle \cdot \left(\langle P_{xx}, P_x \rangle F - \langle P_{xx}, P_y \rangle E \right)$$

$\det ②②$

$$= P_{xy}^T P_{xy} \cdot (EG - F^2)$$

$$- \langle P_x, P_{xy} \rangle \left(\langle P_{xy}, P_x \rangle G - \langle P_{xy}, P_y \rangle F \right)$$

$$+ \langle P_y, P_{xy} \rangle \left(\langle P_{xy}, P_x \rangle F - \langle P_{xy}, P_y \rangle E \right)$$

$\therefore \det ①③ - \det ②②$

$$= \underline{\left(P_{xx}^T P_{yy} - P_{xy}^T P_{xy} \right)} (EG - F^2)$$

↳ We can express this in A.A.A...

$$- \langle P_x, P_{yy} \rangle \cdot \left(\langle P_{xy}, P_x \rangle G - \langle P_{xx}, P_y \rangle F \right)$$

$$+ \langle P_y, P_{yy} \rangle \cdot \left(\langle P_{xx}, P_x \rangle F - \langle P_{xx}, P_y \rangle E \right)$$

$$- \textcircled{0} \cdot (EG - F^2)$$

$$+ \langle P_x, P_{xy} \rangle \left(\langle P_{xy}, P_x \rangle G - \langle P_{xy}, P_y \rangle F \right)$$

$$- \langle P_y, P_{xy} \rangle \left(\langle P_{xy}, P_x \rangle F - \langle P_{xy}, P_y \rangle E \right)$$

So, we can compute

$$\frac{\det \textcircled{1} \det \textcircled{3} - [\det \textcircled{2}]^2}{(EG - F^2)^2} \quad \text{with A.P.A.}$$

Suppose $F = 0$ and $E = G = \lambda$.

$$\partial_x \langle P_x, P_x \rangle = \partial_x \lambda = \langle P_{xx}, P_x \rangle + \langle P_x, P_{xx} \rangle$$

$$\partial_x \langle P_x, P_y \rangle = 0 = \langle P_{xx}, P_y \rangle + \langle P_x, P_{xy} \rangle$$

$$\partial_x \langle P_y, P_y \rangle = \partial_x \lambda = 2 \langle P_{xy}, P_y \rangle$$

$$\partial_y \langle P_y, P_y \rangle = \partial_y \lambda = 2 \langle P_y, P_{yy} \rangle$$

$$\partial_y \langle P_y, P_x \rangle = 0 = \langle P_{yy}, P_x \rangle + \langle P_y, P_{yx} \rangle$$

$$\partial_y \langle P_x, P_x \rangle = \partial_y \lambda = 2 \langle P_{xy}, P_x \rangle$$

$$\langle P_{xx}, P_x \rangle = \langle P_{xy}, P_y \rangle = \tfrac{1}{2} \lambda_x, \quad \langle P_{yy}, P_y \rangle = \langle P_{xy}, P_x \rangle = \tfrac{1}{2} \lambda_y$$

$$\langle P_{yy}, P_x \rangle = -\frac{1}{2} \lambda_x \qquad \langle P_{xx}, P_y \rangle = -\frac{1}{2} \lambda_y$$

$$\frac{1}{2}\lambda_{xx} = \langle P_{xxy}, P_y\rangle + \langle P_{xy}, P_{xy}\rangle$$

$$\frac{1}{2}\lambda_{yy} = \langle P_{xyy}, P_x\rangle + \langle P_{xy}, P_{xy}\rangle \qquad (**)$$

$$0 = \langle P_{xyy}, P_x\rangle + \langle P_{yy}, P_{xy}\rangle + \langle P_{xy}, P_{xy}\rangle + \langle P_y, P_{yxx}\rangle$$

$$\det①③ - \det②②$$

$$= \left(P_{xx}^T P_{yy} - P_{xy}^T P_{xy}\right)\left(EG - F^2\right)$$

$$-\overset{-\frac{1}{2}\lambda_x}{\langle P_x, P_{yy}\rangle}\cdot\left(\overset{\frac{1}{2}\lambda_x}{\langle P_{xx}, P_x\rangle}\overset{\lambda}{G} - \cancel{\langle P_{xx}, P_y\rangle F}\right)$$

$$+\overset{\frac{1}{2}\lambda_y}{\langle P_y, P_{yy}\rangle}\cdot\left(\cancel{\langle P_{xx}, P_x\rangle F} - \overset{-\frac{1}{2}\lambda_y}{\langle P_{xx}, P_y\rangle}\overset{\lambda}{E}\right)$$

$$+\overset{\frac{1}{2}\lambda_y}{\langle P_x, P_{xy}\rangle}\left(\overset{\frac{1}{2}\lambda_y}{\langle P_{xy}, P_x\rangle}\overset{\lambda}{G} - \cancel{\langle P_{xy}, P_y\rangle F}\right)$$

$$-\overset{\frac{1}{2}\lambda_x}{\langle P_y, P_{xy}\rangle}\left(\cancel{\langle P_{xy}, P_x\rangle F} - \overset{\frac{1}{2}\lambda_x}{\langle P_{xy}, P_y\rangle}\overset{\lambda}{E}\right)$$

$$= \left(\underbrace{\langle P_{xx}, P_{yy}\rangle - \langle P_{xy}, P_{xy}\rangle}_{= -\frac{1}{2}(\lambda_{xx}+\lambda_{yy}) \text{ by } (**)}\right)\lambda^2$$

$$+\frac{1}{4}\lambda_x\lambda_x\lambda + \frac{1}{4}\lambda_y\lambda_y\lambda$$

$$+\frac{1}{4}\lambda_y\lambda_y\lambda + \frac{1}{4}\lambda_x\lambda_x\lambda$$

$$= \frac{\lambda^3}{2}\left(-\frac{\lambda_{xx}}{\lambda} - \frac{\lambda_{yy}}{\lambda} + \frac{(\lambda_x)^2}{\lambda} + \frac{(\lambda_y)^2}{\lambda}\right)$$

$$K = \frac{\det①③ - \det②②}{\lambda^2} = -\frac{1}{2\lambda}\left(\partial_x\left(\frac{\lambda_x}{\lambda}\right) + \partial_y\left(\frac{\lambda_y}{\lambda}\right)\right)$$

$$= -\frac{1}{2\lambda}\Delta(\log\lambda) = -\frac{1}{2e^{\sigma}}\Delta\sigma$$

$$\boxed{\frac{4(dx^2+dy^2)}{(1+x^2+y^2)^2}, \quad \lambda = \frac{4}{(1+x^2+y^2)^2}.}$$

$$1 = -\frac{1}{2e^{\sigma}}\Delta\sigma$$

$\Leftrightarrow$ $2e^{\sigma} = -\Delta\sigma$, of course, $\log\left(\dfrac{4}{(1+x^2+y^2)^2}\right)$ is a solution.

$$\left(\because\ \frac{8}{(1+x^2+y^2)^2}\right) = \Delta\left(\log 4 - 2\log(1+x^2+y^2)\right)$$

$$= \partial_x\left(\frac{4x}{1+x^2+y^2}\right) + \partial_y\left(\frac{4y}{(1+x^2+y^2)}\right)$$

$$= \frac{4(1+x^2+y^2)-8x^2+4(1+x^2+y^2)-8y^2}{(1+x^2+y^2)^2}\Bigg)$$

$$\boxed{\text{Is there no other solution?}}$$